

Advanced Machine Learning Syllabus

Course Overview

This course introduces advanced concepts in machine learning.

Topic 1: Introduction to Python for Data Science

Complexity: Low

Prerequisites: None

Description: Basic Python syntax, Pandas, Numpy, and plotting.

Topic 2: Supervised Learning & Regression

Complexity: Medium

Prerequisites: Topic 1

Description: Linear regression, logistic regression, and cost functions.

Topic 3: Neural Networks & Deep Learning

Complexity: High

Prerequisites: Topic 2

Description: Multi-layer perceptrons, backpropagation, and activation functions.

Topic 4: Convolutional Neural Networks (CNNs)

Complexity: High

Prerequisites: Topic 3

Description: Image processing, convolutions, pooling, and ResNet.

Topic 5: Natural Language Processing (NLP)

Complexity: Very High

Prerequisites: Topic 3

Description: Word embeddings, RNNs, and Transformers (BERT/GPT).